Ukraine Has Innovated Naval Warfare

THE MOC

By Mark Temnycky

July 31, 2025

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When Russia launched its full-scale invasion in February 2022, few international experts, news outlets, and organizations expected Ukraine to withstand the Russian invasion. For example, a headline from *Foreign Policy* in read (https://foreignpolicy.com/2022/03/10/ukraine-russia-kyiv-resistance/) that "Ukraine probably cannot hold off Russia forever." The situation was dire, as millions of Ukrainians fled the country, and Russia launched numerous bombardments across Ukraine.

To date, one-fourth of Ukraine's population remains displaced (https://www.cnn.com/2022/03/21/europe/ukraine-russia-conflict-10-million-refugees-intl). Russian forces have killed (https://www.statista.com/statistics/1296924/ukraine-war-casualties-daily/) over 13,000 Ukrainian civilians and destroyed (https://www.rferl.org/a/ukraine-destroyed-cities-russia-war/32454453.html) dozens of cities, villages, and towns.

Despite this destruction, Ukraine and its people continue to fight. They have also achieved several successes during Russia's three-and-a-half-year war. For example, Moscow has lost tens of billions of dollars (https://www.newsweek.com/russia-cost-war-ukraine-tank-losses-military-spending-1887831) in military hardware and equipment, including "almost 90% of their prewar army," two-thirds of their tank force, and thousands of vehicles.

One of Ukraine's most significant achievements during the war, however, has been its campaign in the Black Sea. Ukraine has successfully crippled (https://www.kyivpost.com/post/28412) Russia's Black Sea Fleet. In addition, a report by Business Insider estimated that nearly half (https://www.19fortyfive.com/2025/02/the-russian-navy-could-be-on-the-verge-of-a-big-comeback-or-sink/?_gl=1*rnrogt*_ga*NjQ5NzE1MzUyLjE3NDg0NzY2NDU.*_up*MQ..*_ga_QY074QDH5M*czE3NDg0NzY2NDQkbzEkZzAkdDE3NDg0NzY2NDQkajYwJGwwJGgw) of Russia's naval force on the Black Sea has been damaged or destroyed by Ukraine, a country without a navy (https://www.rferl.org/a/ukraine-navy-black-sea-russia/32826343.html).

How did Ukraine achieve this success? What does Ukraine's Black Sea strategy mean for naval warfare?

For starters, Ukrainian grit and spirit have played a significant role in this war. When Russia's full-scale invasion began, many countries around the world were reluctant (https://www.cfr.org/article/how-much-us-aid-going-ukraine#:~:text=NATO%20allies%20are%20particularly%20wary,to%20strike%20inside%20Russian%20territory.&text=Note:%20Some%20numbers%20are%20approximations.&text=Note:%20Some%20numbers% to send advanced technology and weaponry to Ukraine. The international community had witnessed (https://www.crisisgroup.org/asia/south-asia/afghanistan/afghanistan-three-years-after-taliban-takeover) the fall of Afghanistan to the Taliban a year prior, and watched (https://icct.nl/sites/default/files/2022-12/The-Spoils-of-War-final-1.pdf#:~:text=The%20seemingly%20rushed%20and%20chaotic%20nature%20of,arms%20and%20light%20weapons%20(SALW)%20left%20behind.&text=To%20familiarise%20the%20fighters%20with%20the%20re Taliban militants parade their newly captured Western weapons.. As a result, Western countries such as (https://www.start.umd.edu/publication/us-assistance-ukraine-information-space-intelligence-cyber-and-signaling) the United States and the United Kingdom were hesitant to provide advanced technology and military aid to Ukraine, as some feared that this equipment would fall into Russian hands.

Due to this mindset, many Ukrainians felt that they had to prove themselves to the international community to receive Western defense aid. Millions of Ukrainians understood the task at hand, and they were determined to defend their homeland.

This determination paid off. After the initial Russian onslaught during the spring of 2022, the Ukrainian capital and four-fifths of Ukraine remained intact. The Ukrainian Armed Forces, volunteer battalions, and citizens had successfully defended themselves from Russia's full-scale invasion. The international community then began supplying (https://www.cfr.org/article/how-much-us-aid-going-ukraine#:~:text=NATO%20allies%20are%20particularly%20wary,to%20strike%20inside%20Russian%20territory.&text=Note:%20Some%20numbers%20are%20approximations.&text=Note:%20Some%20numbers% weapons, technology, and other forms of assistance to the Ukrainian military.

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Since then, Ukraine has reclaimed (https://www.reuters.com/world/blinken-says-ukraine-has-taken-back-50-territory-that-russia-seized-2023-07-23/) half of the Russian-occupied territories in southern and eastern parts of the country, and they have decimated (https://www.axios.com/2024/06/04/biden-ukraine-russia-military-decimated) the Russian military. While the war continues, Ukrainians remain resilient (https://carnegieendowment.org/europe/strategic-europe/2025/02/how-ukraine-remains-resilient-three-years-on?lang=en#:~:text=While%20there%20has%20been%20episodic,an%20imperialist%2C%20genocidal%20Russian%20enterprise.&text=Carnegie%20does%20not%20take%20institutional,its%20staff%2C%20or%20its%

Second, innovation and adaptation have been important factors in the war. Take, for example, Ukraine's efforts on the Black Sea. When Russia's full-scale invasion began, the Russians used the Black Sea Fleet to bombard (https://www.marshallcenter.org/en/publications/clock-tower-security-series/strategic-competition-seminar-series-fy24/russia-end-state-battle-black-sea) numerous Ukrainian cities and towns along the coast. In addition, the Moscow established a blockade (https://lordslibrary.parliament.uk/impact-of-ukrainian-port-blockade-on-food-supply-to-developing-countries/) on the Black Sea, which prevented Ukraine from exporting its grain to numerous countries around the world. The Russian blockade also caused a "global food crisis (https://www.consilium.europa.eu/en/infographics/how-the-russian-invasion-of-ukraine-has-further-aggravated-the-global-food-crisis/)" in Africa, the Middle East, and Asia.

The Ukrainian government and military quickly understood that if they targeted Russia's naval forces, this would ease Russian attacks on the Ukrainian coastline and save thousands of lives. In addition, the Ukrainians actively worked to ensure that grain could be sent to numerous countries to avoid a food crisis.

To overcome these issues, the Ukrainians began to utilize radar systems and U.S. intelligence reports to analyze the patterns and movements of the on the Black Sea. After familiarizing themselves with Russia's Black Sea Fleet, the Ukrainians launched a campaign..

The Ukrainians selected the *Moskva*, a guided missile cruiser and the flagship of the Russian Black Sea Fleet. Realizing that the Russian vessel was within range from the Ukrainian coast, the Ukrainian military fired two anti-ship missiles at the *Moskva*. The attack was a success. The ship instantly caught fire, and several hours later, a major symbol of Russia's military sank to the bottom of the Black Sea. This engagement was a significant achievement for Ukraine, as a country without a navy used its land forces to destroy a naval cruiser.

Following the sinking of the *Moskva*, the Russians began to carefully maneuver their fleet on the Black Sea to ensure that another incident like the *Moskva* would not occur. To account for these longer distances, the Ukrainians began redesigning and repurposing (https://news.usni.org/2024/06/13/battles-in-the-black-sea-changing-the-character-of-naval-warfare-experts-say) unmanned surface drones. This effort led to the creation of Ukraine's unmanned surface vessels (USV), known as "Sea Babies". The Ukrainians modified the USVs by integrating explosive payloads and advanced navigation systems, enabling these drones to autonomously and remotely target and damage Russian ships with precise, coordinated attacks (https://www.forbes.com/sites/davidaxe/2024/12/09/ukraines-sea-baby-drone-boats-shoot-back-now). In other words, the Ukrainians transformed the otherwise unarmed USVs into effective naval attack drones capable of overwhelming Russian defenses on the Black Sea.

"[These] defense systems [allow the Ukrainians] to protect themselves from Russian vessels," explained a senior Ukrainian Army Officer. "The Sea Babies and MAGURA V5 drones are equipped with various means that allow the drones to effectively attack Russian targets without putting Ukrainian lives in harm's way."

Ukrainian military turned to drones for several reasons. First, USVs protect Ukrainian servicemen and women. Second, with limited resources (https://nationalinterest.org/blog/buzz/russian-navy-slowly-sinking-black-sea-213806), the Ukrainian military had to tactically evaluate each situation determining whether or not to strike. The use of drones made this decision straightforward, as they can be rebuilt and refurbished if damaged. Third, these unmanned systems gave the Ukrainians the ability to reach Russia's navy from a further distance, as at the time, Ukraine did not have long-range missiles capable of reaching the relocated Russian navy.

After realizing they could reach the Russian vessels with the use of these drones, the Ukrainians began to add explosives and other devices (https://news.usni.org/2024/06/13/battles-in-the-black-sea-changing-the-character-of-naval-warfare-experts-say) to the USVs so that they could target and attack Russian vessels.

This enhancement led to a series of Ukrainian successes. Within a matter of months , these repurposed drones damaged several vessels within the Russian Black Sea Fleet. An influx of Western missiles also allowed (https://www.military.com/daily-news/2024/07/19/how-ukrainians-no-navy-defeated-russias-black-sea-fleet.html) the Ukrainians to target and destroy Russian anti-aircraft systems. Finally, the Ukrainians targeted Russian infrastructure on the Russian-occupied Crimean peninsula, which limited Russia's ability to access ammo deposits and restock its equipment.

Then, in 2023 and 2024, Ukraine's use of drone swarms (https://www.military.com/daily-news/2024/07/19/how-ukrainians-no-navy-defeated-russias-black-sea-fleet.html) altered the balance of naval power in the Black Sea. By developing multiple USVs in coordinated attacks, Ukrainian forces overwhelmed Russian defenses. These swarm tactics not only enabled simultaneous strikes from different vectors, but it also forced Russian commanders to divert resources and develop countermeasures.

Ukrainian forces also equipped some drones with advanced cameras and surveillance technologies (https://www.military.com/daily-news/2024/07/19/how-ukrainians-no-navy-defeated-russias-black-sea-fleet.html). This allowed for wider reconnaissance (https://www.military.com/daily-news/2024/07/19/how-ukrainians-no-navy-defeated-russias-black-sea-fleet.html), helping Ukraine identify vulnerabilities in Russian deployments and map patrol routes. This allowed the Ukrainians to precisely time their attacks on Moscow's Black Sea Fleet.

Furthermore, Ukraine integrated swarm tactics and real-time intelligence to gain a strategic edge. Not only did this degrade the effectiveness of Russia's Black Sea Fleet, but it also demonstrated a new model of naval warfare.

Aside from these strike and reconnaissance missions, the Ukrainians also adapted their unmanned systems to deploy mines (https://en.defence-ua.com/weapon_and_tech/ukraines_sea_baby_naval_drone_can_create_mine_barriers_russian_samum_and_pavel_derzhavin_got_caught-10953.html). By strategically laying these explosive devices near Moscow's Black Sea Fleet, this has further hindered Russia's ability to deploy its navy. s tactic led to the sinking (https://en.defence-ua.com/weapon_and_tech/ukraines_sea_baby_naval_drone_can_create_mine_barriers_russian_samum_and_pavel_derzhavin_got_caught-10953.html) of at least four Russian naval vessels, which were a Ropucha-I-class (https://en.defence-ua.com/weapon_and_tech/ukraines_sea_baby_naval_drone_can_create_mine_barriers_rus) landing ship, the *Sergey Kotov (https://www.rferl.org/a/russia-fleet-destruction-ukraine-drones-war/32689375.html)*, a large Russian patrol ship, the *Saturn (https://en.defence-ua.com/weapon_and_tech/ukraines_sea_baby_naval_drone_can_create_mine_barriers_rus)*, which was a Russian tugboat, and a Serna-class vessel (https://www.navylookout.com/black-sea-battle-how-ukraines-drones-overpowered-the-russian-navy/).

In other words, the Ukrainians developed various approaches to counter Russia's capabilities on the Black Sea. This is known as sea denial (https://www.usni.org/magazines/proceedings/2013/november/getting-sea-control-right), where a smaller naval power restricts (https://www.nationaldefensemagazine.org/articles/2024/9/26/navy-should-establish-hybrid-sea-denial-sea-control-fleet-expert-says) a stronger adversary's freedom of movement and operational ability in key maritime areas without trying to fully control the sea.

By combing drone swarms, unmanned surface vessels carrying explosives, real-time surveillance, and naval mines, Ukraine has made it costly and risky (https://www.navalnews.com/naval-news/2025/03/ukraines-winning-cards-against-russia-in-the-black-sea/) for Russia to use its Black Sea Fleet effectively. Ukraine's approaches (https://www.military.com/daily-news/2024/07/19/how-ukrainians-no-navy-defeated-russias-black-sea-fleet.html) have forced the Russian navy to retreat from forward bases on the Crimean peninsula, restrict Russia's movements on the Black Sea, and limit Russian strikes. Additionally, Ukraine's use of these asymmetric tactics has degraded (https://www.atlanticcouncil.org/blogs/ukrainealert/ukraine-is-shaping-the-future-of-drone-warfare-at-sea-as-well-as-on-land/) Moscow's naval power. These events have marked one of the first practical cases of advanced unmanned systems central to sea denial in contemporary warfare. In other words, Ukraine's strategies have reshaped the naval balance in the Black Sea, enabling the Ukrainians to defend their country.

Finally, Ukraine's successes on the Black Sea have become a powerful symbol of resilience and innovation. Despite Russia's naval superiority and initial control over the Black Sea, Ukraine has turned the tide through the deft use of drone warfare (https://www.atlanticcouncil.org/blogs/ukrainealert/ukraine-is-shaping-the-future-of-drone-warfare-at-sea-as-well-as-on-land/) and advanced missile strikes. This has severely disrupted the operations of Moscow's Black Sea Fleet. Ukraine's victories on the Black Sea have not only reopened trade routes, but these events have challenged (https://neweasterneurope.eu/2025/07/08/why-the-black-sea-is-vital-to-ukraine/) Russia's strategic dominance. These developments have eroded Russia's deterrence narrative, undermining its claims of invincibility.

Overall, repurposing drones with weapons, surveillance equipment, and explosives allowed the Ukrainians to surveil and attack the Russian Black Sea Fleet with limited financial resources. This campaign has led to the destruction of nearly half of Russia's Black Sea Fleet. In addition, these results have saved thousands of lives. Finally, Ukrainian innovation has forced (https://tomorrowsaffairs.com/ukraine-s-innovation-is-raising-the-standards-of-naval-warfare#:~:text=Over%20the%20past%20year%2C%20Ukraine's,for%20any%20future%2Dfocused%20military.) Russia to end its blockade, allowing Kyiv to once again export grain around the world.

Russia's full-scale invasion is far from over, but Ukraine's victories in the Black Sea have inspired its armed forces to press forward in their defense efforts. Ukraine's accomplishments on the Black Sea suggest that innovation and adaptation can help smaller countries achieve success against larger and more aggressive opponents. Military and naval exports will study this Ukrainian campaign for generations to come.

Mark Temnycky *is a nonresident fellow* (https://www.atlanticcouncil.org/expert/mark-temnycky/) *at the Atlantic Council's Eurasia Center and a freelance journalist* (https://wakelet.com/@MTemnycky) *covering Eurasian affairs.*

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Center for Maritime
Strategy
2300 Wilson Boulevard,
Suite 200
Arlington, Virginia
22201

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